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INDICATIONS AND REPORTING FOR BACKSCATTER READING OPERATIONS

Abstract

Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a receiving device may receive an indication of an upcoming backscatter reading operation. Accordingly, the receiving device may adjust a reception parameter of the receiving device in response to the indication. Numerous other aspects are described.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for indications and reporting for backscatter reading operations.

BACKGROUND

[0002] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

[0003] A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the network node to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).

[0004] The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

[0005] Some aspects described herein relate to a method of wireless communication performed by a receiving device. The method may include receiving an indication of an upcoming backscatter reading operation. The method may include adjusting a reception parameter of the receiving device in response to the indication.

[0006] Some aspects described herein relate to a method of wireless communication performed by a receiving device. The method may include receiving an indication of an upcoming backscatter reading operation. The method may include activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0007] Some aspects described herein relate to a method of wireless communication performed by

a receiving device. The method may include measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation. The method may include transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0008] Some aspects described herein relate to a method of wireless communication performed by a transmitting device. The method may include transmitting an indication of an upcoming backscatter reading operation. The method may include transmitting a continuous wave (CW) signal to enable the backscatter reading operation.

[0009] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive an indication of an upcoming backscatter reading operation. The one or more processors may be configured to adjust a reception parameter of the receiving device in response to the indication.

[0010] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive an indication of an upcoming backscatter reading operation. The one or more processors may be configured to activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0011] Some aspects described herein relate to an apparatus for wireless communication at a receiving device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. The one or more processors may be configured to transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0012] Some aspects described herein relate to an apparatus for wireless communication at a transmitting device. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to transmit an indication of an upcoming backscatter reading operation. The one or more processors may be configured to transmit a CW signal to enable the backscatter reading operation.

[0013] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to receive an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to adjust a reception parameter of the receiving device in response to the indication.

[0014] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to receive an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0015] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a receiving device. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the receiving device, may cause the receiving device to transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0016] Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a transmitting device. The set of instructions, when executed by one or more processors of the transmitting device, may cause the transmitting device to transmit an indication of an upcoming backscatter reading operation. The set of instructions, when executed by one or more processors of the transmitting device, may cause the transmitting device to transmit a CW signal to enable the backscatter reading operation.

[0017] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving an indication of an upcoming backscatter reading operation. The apparatus may include means for adjusting a reception parameter of the receiving device in response to the indication.

[0018] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving an indication of an upcoming backscatter reading operation. The apparatus may include means for activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0019] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation. The apparatus may include means for transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0020] Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for transmitting an indication of an upcoming backscatter reading operation. The apparatus may include means for transmitting a CW signal to enable the backscatter reading operation.

[0021] Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, user equipment, base station, network entity, network node, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

[0022] The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.

[0023] While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It

is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.

[0025] FIG. 1 is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.

[0026] FIG. 2 is a diagram illustrating an example of a network node in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.

[0027] FIG. 3 is a diagram illustrating an example disaggregated base station architecture, in accordance with the present disclosure.

[0028] FIGS. 4 and 5 are diagrams illustrating an example of a backscatter reading operation, in accordance with the present disclosure.

[0029] FIG. 6 is a diagram illustrating an example associated with filters for backscatter reading operations, in accordance with the present disclosure.

[0030] FIG. 7 is a diagram illustrating an example associated with reporting for backscatter reading operations, in accordance with the present disclosure.

[0031] FIG. 8 is a diagram illustrating an example associated with performing a backscatter reading operation, in accordance with the present disclosure.

[0032] FIGS. 9, 10, 11, and 12 are diagrams illustrating example processes associated with indications and reporting for backscatter reading operations, in accordance with the present disclosure.

[0033] FIGS. 13 and 14 are diagrams of example apparatuses for wireless communication, in accordance with the present disclosure.

DETAILED DESCRIPTION

[0034] Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0035] Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks,

modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as “elements”). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0036] While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

[0037] FIG. 1 is a diagram illustrating an example of a wireless network **100**, in accordance with the present disclosure. The wireless network **100** may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network **100** may include one or more network nodes **110** (shown as a network node **110a**, a network node **110b**, a network node **110c**, and a network node **110d**), a user equipment (UE) **120** or multiple UEs **120** (shown as a UE **120a**, a UE **120b**, a UE **120c**, a UE **120d**, and a UE **120e**), and/or other entities. A network node **110** is a network node that communicates with UEs **120**. As shown, a network node **110** may include one or more network nodes. For example, a network node **110** may be an aggregated network node, meaning that the aggregated network node is configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node (e.g., within a single device or unit). As another example, a network node **110** may be a disaggregated network node (sometimes referred to as a disaggregated base station), meaning that the network node **110** is configured to utilize a protocol stack that is physically or logically distributed among two or more nodes (such as one or more central units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)).

[0038] In some examples, a network node **110** is or includes a network node that communicates with UEs **120** via a radio access link, such as an RU. In some examples, a network node **110** is or includes a network node that communicates with other network nodes **110** via a fronthaul link or a midhaul link, such as a DU. In some examples, a network node **110** is or includes a network node that communicates with other network nodes **110** via a midhaul link or a core network via a backhaul link, such as a CU. In some examples, a network node **110** (such as an aggregated network node **110** or a disaggregated network node **110**) may include multiple network nodes, such as one or more RUs, one or more CUs, and/or one or more DUs. A network node **110** may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, a transmission reception point (TRP), a DU, an RU, a CU, a mobility element of a network, a core network node, a network element, a network equipment, a RAN node, or a combination thereof. In some examples, the network nodes **110** may be interconnected to one another or to one or more other network nodes **110** in the wireless network **100** through various types of fronthaul, midhaul, and/or backhaul interfaces, such as a direct physical connection, an air interface, or a virtual network, using any suitable transport network.

[0039] In some examples, a network node **110** may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term “cell” can refer to a coverage area of a network node **110** and/or a network node subsystem serving this coverage area, depending on the context in which the term is used. A network node **110** may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **120** with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs **120** with service subscriptions. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs **120** having association with the femto cell (e.g., UEs **120** in a closed subscriber group (CSG)). A network node **110** for a macro cell may be referred to as a macro network node. A network node **110** for a pico cell may be referred to as a pico network node. A network node **110** for a femto cell may be referred to as a femto network node or an in-home

network node. In the example shown in FIG. 1, the network node **110a** may be a macro network node for a macro cell **102a**, the network node **110b** may be a pico network node for a pico cell **102b**, and the network node **110c** may be a femto network node for a femto cell **102c**. A network node may support one or multiple (e.g., three) cells. In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a network node **110** that is mobile (e.g., a mobile network node).

[0040] In some aspects, the term “base station” or “network node” may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, or one or more components thereof. For example, in some aspects, “base station” or “network node” may refer to a CU, a DU, an RU, a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some aspects, the term “base station” or “network node” may refer to one device configured to perform one or more functions, such as those described herein in connection with the network node **110**. In some aspects, the term “base station” or “network node” may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a quantity of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term “base station” or “network node” may refer to any one or more of those different devices. In some aspects, the term “base station” or “network node” may refer to one or more virtual base stations or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term “base station” or “network node” may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.

[0041] The wireless network **100** may include one or more relay stations. A relay station is a network node that can receive a transmission of data from an upstream node (e.g., a network node **110** or a UE **120**) and send a transmission of the data to a downstream node (e.g., a UE **120** or a network node **110**). A relay station may be a UE **120** that can relay transmissions for other UEs **120**. In the example shown in FIG. 1, the network node **110d** (e.g., a relay network node) may communicate with the network node **110a** (e.g., a macro network node) and the UE **120d** in order to facilitate communication between the network node **110a** and the UE **120d**. A network node **110** that relays communications may be referred to as a relay station, a relay base station, a relay network node, a relay node, a relay, or the like.

[0042] The wireless network **100** may be a heterogeneous network that includes network nodes **110** of different types, such as macro network nodes, pico network nodes, femto network nodes, relay network nodes, or the like. These different types of network nodes **110** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **100**. For example, macro network nodes may have a high transmit power level (e.g., 5 to 40 watts) whereas pico network nodes, femto network nodes, and relay network nodes may have lower transmit power levels (e.g., 0.1 to 2 watts).

[0043] A network controller **130** may couple to or communicate with a set of network nodes **110** and may provide coordination and control for these network nodes **110**. The network controller **130** may communicate with the network nodes **110** via a backhaul communication link or a midhaul communication link. The network nodes **110** may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. In some aspects, the network controller **130** may be a CU or a core network device, or may include a CU or a core network device.

[0044] The UEs **120** may be dispersed throughout the wireless network **100**, and each UE **120** may be stationary or mobile. A UE **120** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **120** may be a cellular phone (e.g., a smart phone), a

personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, a UE function of a network node, and/or any other suitable device that is configured to communicate via a wireless or wired medium.

[0045] Some UEs **120** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network node, another device (e.g., a remote device), or some other entity. Some UEs **120** may be considered Internet-of-Things (IOT) devices, and/or may be implemented as NB-IOT (narrowband IoT) devices. Some UEs **120** may be considered a Customer Premises Equipment. A UE **120** may be included inside a housing that houses components of the UE **120**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

[0046] In general, any number of wireless networks **100** may be deployed in a given geographic area. Each wireless network **100** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

[0047] In some examples, two or more UEs **120** (e.g., shown as UE **120a** and UE **120e**) may communicate directly using one or more sidelink channels (e.g., without using a network node **110** as an intermediary to communicate with one another). For example, the UEs **120** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **120** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the network node **110**.

[0048] Devices of the wireless network **100** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **100** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0049] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend

features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0050] With the above examples in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

[0051] In some aspects, the UE **120** may include a communication manager **140**. As described in more detail elsewhere herein, the communication manager **140** may receive an indication of an upcoming backscatter reading operation and adjust a reception parameter of the UE **120** in response to the indication. Additionally, or alternatively, the communication manager **140** may receive an indication of an upcoming backscatter reading operation and activate a low-power mode for one or more hardware components of the UE **120** in response to the indication. Additionally, or alternatively, the communication manager **140** may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation and transmit a report indicating a set of measurements of the set of beamformed reference signals. Additionally, or alternatively, the communication manager **140** may perform one or more other operations described herein.

[0052] In some aspects, the network node **110** may include a communication manager **150**. As described in more detail elsewhere herein, the communication manager **150** may transmit an indication of an upcoming backscatter reading operation; and transmit a continuous wave (CW) signal to enable the backscatter reading operation. Additionally, or alternatively, the communication manager **150** may perform one or more other operations described herein.

[0053] As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

[0054] FIG. **2** is a diagram illustrating an example **200** of a network node **110** in communication with a UE **120** in a wireless network **100**, in accordance with the present disclosure. The network node **110** may be equipped with a set of antennas **234a** through **234t**, such as T antennas ($T \geq 1$). The UE **120** may be equipped with a set of antennas **252a** through **252r**, such as R antennas ($R \geq 1$). The network node **110** of example **200** includes one or more radio frequency components, such as antennas **234** and a modem **254**. In some examples, a network node **110** may include an interface, a communication component, or another component that facilitates communication with the UE **120** or another network node. Some network nodes **110** may not include radio frequency components that facilitate direct communication with the UE **120**, such as one or more CUs, or one or more DUs.

[0055] At the network node **110**, a transmit processor **220** may receive data, from a data source **212**, intended for the UE **120** (or a set of UEs **120**). The transmit processor **220** may select one or more modulation and coding schemes (MCSs) for the UE **120** based at least in part on one or more channel quality indicators (CQIs) received from that UE **120**. The network node **110** may process (e.g., encode and modulate) the data for the UE **120** based at least in part on the MCS(s) selected for the UE **120** and may provide data symbols for the UE **120**. The transmit processor **220** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor **220** may generate reference symbols for

reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems **232** (e.g., T modems), shown as modems **232a** through **232t**. For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem **232**. Each modem **232** may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem **232** may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems **232a** through **232t** may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas **234** (e.g., T antennas), shown as antennas **234a** through **234t**.

[0056] At the UE **120**, a set of antennas **252** (shown as antennas **252a** through **252r**) may receive the downlink signals from the network node **110** and/or other network nodes **110** and may provide a set of received signals (e.g., R received signals) to a set of modems **254** (e.g., R modems), shown as modems **254a** through **254r**. For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a modem **254**. Each modem **254** may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem **254** may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector **256** may obtain received symbols from the modems **254**, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor **258** may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE **120** to a data sink **260**, and may provide decoded control information and system information to a controller/processor **280**. The term “controller/processor” may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples. In some examples, one or more components of the UE **120** may be included in a housing **284**.

[0057] The network controller **130** may include a communication unit **294**, a controller/processor **290**, and a memory **292**. The network controller **130** may include, for example, one or more devices in a core network. The network controller **130** may communicate with the network node **110** via the communication unit **294**.

[0058] One or more antennas (e.g., antennas **234a** through **234t** and/or antennas **252a** through **252r**) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. 2.

[0059] On the uplink, at the UE **120**, a transmit processor **264** may receive and process data from a data source **262** and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor **280**. The transmit processor **264** may generate reference symbols for one or more reference signals. The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modems **254** (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node **110**. In some examples, the

modem **254** of the UE **120** may include a modulator and a demodulator. In some examples, the UE **120** includes a transceiver. The transceiver may include any combination of the antenna(s) **252**, the modem(s) **254**, the MIMO detector **256**, the receive processor **258**, the transmit processor **264**, and/or the TX MIMO processor **266**. The transceiver may be used by a processor (e.g., the controller/processor **280**) and the memory **282** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **6-14**).

[0060] At the network node **110**, the uplink signals from UE **120** and/or other UEs may be received by the antennas **234**, processed by the modem **232** (e.g., a demodulator component, shown as DEMOD, of the modem **232**), detected by a MIMO detector **236** if applicable, and further processed by a receive processor **238** to obtain decoded data and control information sent by the UE **120**. The receive processor **238** may provide the decoded data to a data sink **239** and provide the decoded control information to the controller/processor **240**. The network node **110** may include a communication unit **244** and may communicate with the network controller **130** via the communication unit **244**. The network node **110** may include a scheduler **246** to schedule one or more UEs **120** for downlink and/or uplink communications. In some examples, the modem **232** of the network node **110** may include a modulator and a demodulator. In some examples, the network node **110** includes a transceiver. The transceiver may include any combination of the antenna(s) **234**, the modem(s) **232**, the MIMO detector **236**, the receive processor **238**, the transmit processor **220**, and/or the TX MIMO processor **230**. The transceiver may be used by a processor (e.g., the controller/processor **240**) and the memory **242** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **6-14**).

[0061] The controller/processor **240** of the network node **110**, the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform one or more techniques associated with indications and reporting for backscatter reading operations, as described in more detail elsewhere herein. For example, the controller/processor **240** of the network node **110**, the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform or direct operations of, for example, process **900** of FIG. **9**, process **1000** of FIG. **10**, process **1100** of FIG. **11**, process **1200** of FIG. **12**, and/or other processes as described herein. The memory **242** and the memory **282** may store data and program codes for the network node **110** and the UE **120**, respectively. In some examples, the memory **242** and/or the memory **282** may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network node **110** and/or the UE **120**, may cause the one or more processors, the UE **120**, and/or the network node **110** to perform or direct operations of, for example, process **900** of FIG. **9**, process **1000** of FIG. **10**, process **1100** of FIG. **11**, process **1200** of FIG. **12**, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples. In some aspects, the transmitting device described herein is the UE **120**, is included in the UE **120**, includes one or more components of the UE **120** shown in FIG. **2**, is the network node **110**, is included in the network node **110**, or includes one or more components of the network node **110** shown in FIG. **2**. In some aspects, the receiving device described herein is the UE **120**, is included in the UE **120**, or includes one or more components of the UE **120** shown in FIG. **2**.

[0062] In some aspects, a receiving device (e.g., a UE **120**) may include means for receiving an indication of an upcoming backscatter reading operation; and/or means for adjusting a reception parameter of the receiving device in response to the indication. Additionally, or alternatively, the receiving device may include means for receiving an indication of an upcoming backscatter reading operation; and/or means for activating a low-power mode for one or more hardware components of the receiving device in response to the indication. Additionally, or alternatively, the receiving

device may include means for measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation; and/or means for transmitting a report indicating a set of measurements of the set of beamformed reference signals. In some aspects, the means for the receiving device to perform operations described herein may include, for example, one or more of communication manager **140**, antenna **252**, modem **254**, MIMO detector **256**, receive processor **258**, transmit processor **264**, TX MIMO processor **266**, controller/processor **280**, or memory **282**.

[0063] In some aspects, a transmitting device (e.g., a UE **120** and/or a network node **110**) may include means for transmitting an indication of an upcoming backscatter reading operation; and/or means for transmitting a CW signal to enable the backscatter reading operation. In some aspects, the means for the transmitting device to perform operations described herein may include, for example, one or more of communication manager **150**, transmit processor **220**, TX MIMO processor **230**, modem **232**, antenna **234**, MIMO detector **236**, receive processor **238**, controller/processor **240**, memory **242**, or scheduler **246**. Alternatively, the means for the transmitting device to perform operations described herein may include, for example, one or more of communication manager **140**, antenna **252**, modem **254**, MIMO detector **256**, receive processor **258**, transmit processor **264**, TX MIMO processor **266**, controller/processor **280**, or memory **282**.

[0064] While blocks in FIG. 2 are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware, software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor **264**, the receive processor **258**, and/or the TX MIMO processor **266** may be performed by or under the control of the controller/processor **280**.

[0065] As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

[0066] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, a base station, or a network equipment may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), an evolved NB (eNB), an NR BS, a 5G NB, an access point (AP), a TRP, or a cell, among other examples), or one or more units (or one or more components) performing base station functionality, may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station. “Network entity” or “network node” may refer to a disaggregated base station, or to one or more units of a disaggregated base station (such as one or more CUs, one or more DUs, one or more RUs, or a combination thereof).

[0067] An aggregated base station (e.g., an aggregated network node) may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node (e.g., within a single device or unit). A disaggregated base station (e.g., a disaggregated network node) may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some examples, a CU may be implemented within a network node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other network nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, such as a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU), among other examples.

[0068] Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)) to facilitate scaling of communication systems by separating base station

functionality into one or more units that can be individually deployed. A disaggregated base station may include functionality implemented across two or more units at various physical locations, as well as functionality implemented for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station can be configured for wired or wireless communication with at least one other unit of the disaggregated base station.

[0069] FIG. 3 is a diagram illustrating an example disaggregated base station architecture 300, in accordance with the present disclosure. The disaggregated base station architecture 300 may include a CU 310 that can communicate directly with a core network 320 via a backhaul link, or indirectly with the core network 320 through one or more disaggregated control units (such as a Near-RT RIC 325 via an E2 link, or a Non-RT RIC 315 associated with a Service Management and Orchestration (SMO) Framework 305, or both). A CU 310 may communicate with one or more DUs 330 via respective midhaul links, such as through F1 interfaces. Each of the DUs 330 may communicate with one or more RUs 340 via respective fronthaul links. Each of the RUs 340 may communicate with one or more UEs 120 via respective radio frequency (RF) access links. In some implementations, a UE 120 may be simultaneously served by multiple RUs 340.

[0070] Each of the units, including the CUs 310, the DUs 330, the RUs 340, as well as the Near-RT RICs 325, the Non-RT RICs 315, and the SMO Framework 305, may include one or more interfaces or be coupled with one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to one or multiple communication interfaces of the respective unit, can be configured to communicate with one or more of the other units via the transmission medium. In some examples, each of the units can include a wired interface, configured to receive or transmit signals over a wired transmission medium to one or more of the other units, and a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units. In some aspects, the CU 310 may host one or more higher layer control

[0071] functions. Such control functions can include radio resource control (RRC) functions, packet data convergence protocol (PDCP) functions, or service data adaptation protocol (SDAP) functions, among other examples. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 310. The CU 310 may be configured to handle user plane functionality (for example, Central Unit-User Plane (CU-UP) functionality), control plane functionality (for example, Central Unit-Control Plane (CU-CP) functionality), or a combination thereof. In some implementations, the CU 310 can be logically split into one or more CU-UP units and one or more CU-CP units. A CU-UP unit can communicate bidirectionally with a CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 310 can be implemented to communicate with a DU 330, as necessary, for network control and signaling.

[0072] Each DU 330 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 340. In some aspects, the DU 330 may host one or more of a radio link control (RLC) layer, a MAC layer, and one or more high physical (PHY) layers depending, at least in part, on a functional split, such as a functional split defined by the 3GPP. In some aspects, the one or more high PHY layers may be implemented by one or more modules for forward error correction (FEC) encoding and decoding, scrambling, and modulation and demodulation, among other examples. In some aspects, the DU 330 may further host one or more low PHY layers, such as implemented by one or more modules for a fast Fourier transform (FFT), an inverse FFT (iFFT), digital beamforming, or physical random access channel (PRACH) extraction and filtering, among other examples. Each layer (which also may be referred to as a module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 330, or with the control functions hosted by the CU 310.

[0073] Each RU **340** may implement lower-layer functionality. In some deployments, an RU **340**, controlled by a DU **330**, may correspond to a logical node that hosts RF processing functions or low-PHY layer functions, such as performing an FFT, performing an iFFT, digital beamforming, or PRACH extraction and filtering, among other examples, based on a functional split (for example, a functional split defined by the 3GPP), such as a lower layer functional split. In such an architecture, each RU **340** can be operated to handle over the air (OTA) communication with one or more UEs **120**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **340** can be controlled by the corresponding DU **330**. In some scenarios, this configuration can enable each DU **330** and the CU **310** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0074] The SMO Framework **305** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **305** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements, which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **305** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) platform **390**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **310**, DUs **330**, RUs **340**, non-RT RICs **315**, and Near-RT RICs **325**. In some implementations, the SMO Framework **305** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **311**, via an O1 interface. Additionally, in some implementations, the SMO Framework **305** can communicate directly with each of one or more RUs **340** via a respective O1 interface. The SMO Framework **305** also may include a Non-RT RIC **315** configured to support functionality of the SMO Framework **305**.

[0075] The Non-RT RIC **315** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **325**. The Non-RT RIC **315** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **325**. The Near-RT RIC **325** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **310**, one or more DUs **330**, or both, as well as an O-eNB, with the Near-RT RIC **325**.

[0076] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **325**, the Non-RT RIC **315** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **325** and may be received at the SMO Framework **305** or the Non-RT RIC **315** from non-network data sources or from network functions. In some examples, the Non-RT RIC **315** or the Near-RT RIC **325** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **315** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **305** (such as reconfiguration via an O1 interface) or via creation of RAN management policies (such as A1 interface policies).

[0077] As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

[0078] FIG. 4 is a diagram illustrating an example **400** of a backscatter reading operation, in accordance with the present disclosure. Example **400** includes a transmitting (TX) device **401** (e.g., a UE **120** or a network node **110**), a receiving (RX) device **403** (e.g., a UE **120**), and a tag **405** (or another passive or semi-active device that backscatters signal from the TX device **401**).

[0079] As shown in FIG. 4, the TX device **401** may communicate with the RX device **403** over a

direct link. For example, the TX device **401** may communicate over a PC5 interface (e.g., associated with a sidelink channel) with the RX device **403** or over a Uu interface with the RX device **403**. In example **400**, a signal over the direct link is represented by $hD1D2(n)$.

[0080] Additionally, or alternatively, as shown in FIG. **4**, the TX device **401** may transmit a signal (e.g., a CW signal) to the tag **405**. In example **400**, the CW signal is represented by $hDIT(n)$. The tag **405** may therefore backscatter the CW signal, resulting in a backscattered signal arriving at the RX device **403**. In example **400**, the signal transmitted by the tag **405** is represented by $hTD2(n)$.

[0081] Accordingly, the RX device **403** may decode the backscattered signal to receive information from the tag **405**. Alternatively, the RX device **403** may decode a superposition of the backscattered signal from the tag **405** and a signal over the direct link from the TX device **401**.

[0082] FIG. **5** is a diagram illustrating an example **500** of the signals for a backscatter reading operation (e.g., the backscatter reading operation of FIG. **4**), in accordance with the present disclosure. FIG. **5** shows an example graph of power for a CW signal **501** received at the RX device **403** (e.g., represented by $hD1D2(n)$) over time. FIG. **5** further shows an example graph of power for the backscattered signal **503** arriving at the RX device **403**. In example **500**, the signal arriving at the RX device **403** is a combination of the CW signal received at the tag **405** (e.g., represented by $h1IT(n)$), the backscattered signal transmitted by the tag **405** (e.g., represented by $hTD2(n)$), an information signal applied by the tag **405** (e.g., represented by $s(n)$), and a power loss factor due to absorption of the CW signal at the tag **405** (e.g., represented by of). Accordingly, as shown in FIG. **5**, the RX device **403** may decode the combined signal **505** that is received, which is a combination of the CW signal **501** received at the RX device **403** (e.g., represented by $hD1D2(n)$) and the backscattered signal arriving at the RX device **403**.

[0083] In order to activate a tag, a TX device generally uses a CW signal with a much larger transmit power than other signals. However, decoding a combined signal is particularly difficult when the CW signal dominates receiving hardware at an RX device (e.g., an analog-to-digital converter (ADC), a low noise amplifier (LNA), an automatic gain control (AGC) loop, and/or components in an RF front end (RFFE), among other examples). Additionally, the CW signal may be strong enough to damage receiving hardware at the RX device and/or at other devices (e.g., other UEs) nearby the RX device.

[0084] Some techniques and apparatuses described herein enable a TX device (e.g., TX device **401**) to indicate an upcoming backscatter reading operation. Accordingly, an RX device (e.g., RX device **403**) may adjust a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device **403** conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device **401**. Additionally, a nearby device may activate a low-power mode for one or more hardware components during the backscatter reading operation. As a result, the nearby device prevents possible hardware failure during the backscatter reading operation.

[0085] Additionally, or alternatively, some techniques and apparatuses described herein enable the TX device **401** to request measurements associated with different beamforming directions (also referred to as “beams”). Accordingly, the RX device **403** (and/or a nearby device) may report measurements of the beams. As a result, the TX device **401** may select a beam for a CW signal that minimizes chances of damage to hardware components of the RX device **403** and any nearby devices.

[0086] As indicated above, FIGS. **4** and **5** are provided as an example. Other examples may differ from what is described with respect to FIGS. **4** and **5**.

[0087] FIG. **6** is a diagram illustrating an example **600** associated with filters for backscatter reading operations, in accordance with the present disclosure. In example **600**, a TX device **401** (e.g., a UE **120** or a network node **110**) may indicate an upcoming backscatter reading operation to an RX device **403** (e.g., a UE **120**). The indication may be a layer 1 (L1) indication (e.g., downlink control information (DCI) or sidelink control information (SCI)), a layer 2 (L2) indication (e.g., a

medium access control (MAC) control element (MAC-CE) or a sidelink MAC-CE), and/or a layer 3 (L3) indication (e.g., an RRC message or a sidelink RRC message). Although described in connection with the TX device **401** and the RX device **403** communicating over a Uu interface or a PC5 interface, the TX device **401** and the RX device **403** may alternatively communicate over a different interface that allows the TX device **401** to read a tag (e.g., tag **405**) in coordination with the RX device **403**.

[0088] Accordingly, the TX device **401** may transmit a CW signal that has a power **601**, as shown in FIG. 6, in order to initiate the backscatter reading operation. Therefore, the RX device **403** may activate an analog stopband filter based on one or more CW frequencies. For example, the TX device **401** may indicate the one or more CW frequencies when indicating the upcoming backscatter reading operation.

[0089] The backscatter signal to be decoded by the RX device **403** may have a power **603a** in a first set of frequencies and a power **603b** in a second set of frequencies, as shown in FIG. 6. Additionally, or alternatively, the RX device **403** may activate an analog passband filter based on one or more desired frequencies (e.g., the first set of frequencies and the second set of frequencies). For example, the TX device **401** may indicate the one or more desired frequencies when indicating the upcoming backscatter reading operation.

[0090] The RX device **403** may transmit a capability message to the TX device **401** that indicates settings associated with an analog passband of the RX device **403** and/or settings associated with an analog stopband of the RX device **403**. For example, the RX device **403** may indicate the bandwidths and/or the filtering values available for the analog stopband and/or the analog passband. Accordingly, the TX device **401** may indicate the bandwidths and/or the filtering values to use when indicating the upcoming backscatter reading operation.

[0091] By using techniques as described in connection with FIG. 6, the TX device **401** to indicate an upcoming backscatter reading operation. Accordingly, the RX device **403** adjusts a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device **403** conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device **401**.

[0092] As indicated above, FIG. 6 is provided as an example. Other examples may differ from what is described with respect to FIG. 6.

[0093] FIG. 7 is a diagram illustrating an example **700** associated with reporting for backscatter reading operations, in accordance with the present disclosure. In example **700**, a TX device **401** (e.g., a UE **120** or a network node **110**) may indicate an upcoming backscatter reading operation to an RX device **403** (e.g., a UE **120**). The indication may be an L1 indication, an L2 indication, and/or an L3 indication. Although described in connection with the TX device **401** and the RX device **403** communicating over a Uu interface or a PC5 interface, the TX device **401** and the RX device **403** may alternatively communicate over a different interface that allows the TX device **401** to read a tag (e.g., tag **405**) in coordination with the RX device **403**.

[0094] Accordingly, the RX device **403** (or a nearby device) may perform one or more measurements (e.g., on one or more reference signals from the TX device **401**) and transmit a report to the TX device **401** based on the measurement(s). For example, the report may indicate a power delta **701** between a CW signal (e.g., a reference signal, such as a channel state information reference signal (CSI-RS) or a sidelink CSI-RS (SL CSI-RS)) and a backscatter signal to be decoded (e.g., caused by the reference signal). Accordingly, the TX device **401** may reduce a transmit power associated with the CW signal when the RX device **403** indicates that the CW signal is overwhelming the hardware of the RX device **403** (e.g., based on the power delta **701** and/or another measurement value in the report). Conversely, the TX device **401** may increase a transmit power associated with the CW signal when the RX device **403** indicates that the backscatter signal cannot be detected or decoded (e.g., based on the power delta **701** and/or another measurement value in the report).

[0095] Although shown with respect to the backscatter signal, the power delta **701** may be measured with respect to other reference signals (or a direct signal from the TX device **401**). Similarly, although described with respect to a delta, the report may additionally or alternatively include a ratio between a power of the CW signal and a power of the backscatter signal (or another reference signal or a direct signal from the TX device **401**) and/or a headroom value (e.g., based on any power delta described herein).

[0096] Additionally, or alternatively, the RX device **403** may transmit a power control command to the RX device **403**. For example, the RX device **403** may indicate to reduce a transmit power associated with the CW signal when the CW signal is overwhelming the hardware of the RX device **403**. Conversely, the RX device **403** may indicate to increase a transmit power associated with the CW signal when the backscatter signal cannot be detected or decoded.

[0097] In some aspects, the TX device **401** may additionally or alternatively transmit a configuration associated with a set of beamformed reference signals associated with the upcoming backscatter reading operation. Accordingly, the RX device **403** (and/or a nearby device) may perform measurements on the set of beamformed reference signals and report RSRP (or sidelink RSRP) values, RSRQ (or sidelink RSRQ) values, and/or signal-to-interference-plus-noise ratio (SINR) (or sidelink SINR) values to the TX device **401**. Accordingly, the TX device **401** may select a beam associated with a lowest measurement value (e.g., at the RX device **403** and/or at a nearby device) in order to reduce chances of damage to hardware components of the RX device **403** and any nearby devices.

[0098] By using techniques as described in connection with FIG. 7, the TX device **401** requests measurements associated with the backscatter reading operation. As a result, the TX device **401** may apply power control for the CW signal that minimizes chances of damage to hardware components of the RX device **403** and any nearby devices.

[0099] As indicated above, FIG. 7 is provided as an example. Other examples may differ from what is described with respect to FIG. 7.

[0100] FIG. 8 is a diagram illustrating an example **800** associated with performing a backscatter reading operation, in accordance with the present disclosure. As shown in FIG. 8, a TX device **401** may perform a backscatter reading operation using a tag **405** and an RX device **403**. A nearby device **801** (e.g., a UE **120** or a network node **110**) may be near the tag **405** and/or the RX device **403**. Although described in connection with the TX device **401** and the RX device **403** communicating over a Uu interface or a PC5 interface, the TX device **401** and the RX device **403** may alternatively communicate over a different interface that allows the TX device **401** to read the tag **405** in coordination with the RX device **403**.

[0101] In some aspects, the TX device **401** may select the RX device **403** to perform the backscatter reading operation based on a location of the RX device **403** relative to one or more locations of one or more devices near the receiving device, such as the nearby device **801**. Accordingly, the TX device **401** may select an RX device **403** that is furthest from a set of nearby devices (e.g., based on median distance, a clustering factor, and/or another similar measure) in order to reduce chances of damage to as many of the set of nearby device as possible.

[0102] As shown by reference number **805**, the TX device **401** may transmit an indication of an upcoming backscatter reading operation. For example, the indication may be an L1 indication, an L2 indication, and/or an L3 indication. The RX device **403** may receive the indication along with the nearby device **801**.

[0103] Accordingly, as shown by reference number **810**, the RX device **403** may adjust a reception parameter in response to the indication. For example, the RX device **403** may enable an analog stopband filter and/or an analog passband filter, as described in connection with FIG. 6. Additionally, or alternatively, the RX device **403** may change an ADC to operate in a different (e.g., higher) dynamic range in response to the indication.

[0104] Further, as shown by reference number **815**, the nearby device **801** may activate a low-

power mode for one or more hardware components of the nearby device **801** in response to the indication. As used herein, a “low-power mode” refers to a mode in which a hardware component is not supplied power or is supplied power below a usual operating power. The nearby device **801** may activate the low-power mode for an LNA, an AGC loop, an RFFE component, and/or another hardware component associated with receiving and processing wireless signals. Additionally, in some aspects, the nearby device **801** may enable an analog stopband filter and/or an analog passband filter, as described in connection with FIG. 6.

[0105] Accordingly, as shown by reference number **820**, the TX device **401** may transmit a CW signal to enable the backscatter reading operation. For example, the CW signal may be beamformed based on a report from the RX device **403** and/or the nearby device **801**, as described in connection with FIG. 7. Additionally, the CW signal may have a transmit power determined based on a report from the RX device **403** and/or the nearby device **801**, as described in connection with FIG. 7.

[0106] In some aspects, data programmed into the tag **405** may be associated with a priority and/or a quality-of-service (QoS). Accordingly, the TX device **401** may schedule the CW signal based on the priority and/or the QoS associated with the tag **405**. Therefore, data that the TX device **401** is to transmit to the RX device **403** (and/or the nearby device **801**) may be transmitted before or after the CW signal, depending on the priorities and/or the QoSs associated with the data to be transmitted and the tag **405**.

[0107] As shown by reference number **825**, the RX device **403** may receive a backscattered signal (e.g., from the tag **405**) using the adjusted reception parameter. Accordingly, the RX device **403** may decode the data programmed into the tag **405** (and included in the backscattered signal, as described in connection with FIGS. 4 and 5). As shown by reference number **830**, the RX device **403** may relay the decoded data from the tag **405** to the TX device **401**.

[0108] By using techniques as described in connection with FIG. 8, the TX device **401** indicates an upcoming backscatter reading operation. Accordingly, the RX device **403** may adjust a receiving parameter to improve accuracy when decoding the combined signal. As a result, the RX device **403** conserves power and processing resources that would otherwise have been wasted on a failed decoding attempt and on a subsequent retransmission by the TX device **401**. Additionally, the nearby device **801** may activate a low-power mode for one or more hardware components during the backscatter reading operation. As a result, the nearby device **801** prevents possible hardware failure during the backscatter reading operation.

[0109] As indicated above, FIG. 8 is provided as an example. Other examples may differ from what is described with respect to FIG. 8.

[0110] FIG. 9 is a diagram illustrating an example process **900** performed, for example, by a receiving device, in accordance with the present disclosure. Example process **900** is an example where the receiving device (e.g., RX device **403** and/or apparatus **1300** of FIG. 13) performs operations associated with indications for backscatter reading operations.

[0111] As shown in FIG. 9, in some aspects, process **900** may include receiving an indication of an upcoming backscatter reading operation (block **910**). For example, the receiving device (e.g., using communication manager **140** and/or reception component **1302**, depicted in FIG. 13) may receive an indication of an upcoming backscatter reading operation, as described herein.

[0112] As further shown in FIG. 9, in some aspects, process **900** may include adjusting a reception parameter of the receiving device in response to the indication (block **920**). For example, the receiving device (e.g., using communication manager **140** and/or adjustment component **1308**, depicted in FIG. 13) may adjust a reception parameter of the receiving device in response to the indication, as described herein.

[0113] Process **900** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0114] In a first aspect, the indication is included in DCI, a MAC-CE, or an RRC message.

[0115] In a second aspect, alone or in combination with the first aspect, the indication is included in SCI, a sidelink MAC-CE, or a sidelink RRC message.

[0116] In a third aspect, alone or in combination with one or more of the first and second aspects, adjusting the reception parameter includes enabling an analog passband filter based on one or more desired frequencies, enabling an analog stopband filter based on one or more continuous wave frequencies, or a combination thereof.

[0117] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process **900** includes transmitting (e.g., using communication manager **140** and/or transmission component **1304**, depicted in FIG. **13**) a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0118] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, adjusting the reception parameter includes adjusting a dynamic range associated with an ADC of the receiving device.

[0119] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process **900** includes transmitting (e.g., using communication manager **140** and/or transmission component **1304**) a capability message indicating a dynamic range associated with an ADC of the receiving device, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0120] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, process **900** includes receiving (e.g., using communication manager **140** and/or reception component **1302**) a backscattered signal using the adjusted reception parameter, such that the backscattered signal is received based on at least one of an associated priority or an associated QoS.

[0121] Although FIG. **9** shows example blocks of process **900**, in some aspects, process **900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **9**. Additionally, or alternatively, two or more of the blocks of process **900** may be performed in parallel.

[0122] FIG. **10** is a diagram illustrating an example process **1000** performed, for example, by a receiving device, in accordance with the present disclosure. Example process **1000** is an example where the receiving device (e.g., RX device **403** and/or apparatus **1300** of FIG. **13**) performs operations associated with indications for backscatter reading operations.

[0123] As shown in FIG. **10**, in some aspects, process **1000** may include receiving an indication of an upcoming backscatter reading operation (block **1010**). For example, the receiving device (e.g., using communication manager **140** and/or reception component **1302**, depicted in FIG. **13**) may receive an indication of an upcoming backscatter reading operation, as described herein.

[0124] As further shown in FIG. **10**, in some aspects, process **1000** may include

[0125] activating a low-power mode for one or more hardware components of the receiving device in response to the indication (block **1020**). For example, the receiving device (e.g., using communication manager **140** and/or power management component **1310**, depicted in FIG. **13**) may activate a low-power mode for one or more hardware components of the receiving device in response to the indication, as described herein.

[0126] Process **1000** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0127] In a first aspect, the indication is included in DCI, a MAC-CE, or an RRC message.

[0128] In a second aspect, alone or in combination with the first aspect, the indication is included in SCI, a sidelink MAC-CE, or a sidelink RRC message.

[0129] In a third aspect, alone or in combination with one or more of the first and second aspects, the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

[0130] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process **1000** includes enabling (e.g., using communication manager **140** and/or adjustment component **1308**, depicted in FIG. **13**) an analog passband filter based on one or more desired frequencies, enabling (e.g., using communication manager **140** and/or adjustment component **1308**) an analog stopband filter based on one or more CW frequencies, or a combination thereof.

[0131] Although FIG. **10** shows example blocks of process **1000**, in some aspects, process **1000** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **10**. Additionally, or alternatively, two or more of the blocks of process **1000** may be performed in parallel.

[0132] FIG. **11** is a diagram illustrating an example process **1100** performed, for example, by a receiving device, in accordance with the present disclosure. Example process **1100** is an example where the receiving device (e.g., RX device **403** and/or apparatus **1300** of FIG. **13**) performs operations associated with reporting for backscatter reading operations.

[0133] As shown in FIG. **11**, in some aspects, process **1100** may include measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation (block **1110**). For example, the receiving device (e.g., using communication manager **140** and/or measurement component **1312**, depicted in FIG. **13**) may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation, as described herein.

[0134] As further shown in FIG. **11**, in some aspects, process **1100** may include transmitting a report indicating a set of measurements of the set of beamformed reference signals (block **1120**). For example, the receiving device (e.g., using communication manager **140** and/or transmission component **1304**, depicted in FIG. **13**) may transmit a report indicating a set of measurements of the set of beamformed reference signals, as described herein.

[0135] Process **1100** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0136] In a first aspect, the set of measurements includes one or more RSRP values, one or more RSRQ values, one or more SINR values, or a combination thereof.

[0137] In a second aspect, alone or in combination with the first aspect, the report indicates a ratio between a direct signal and a continuous wave signal.

[0138] In a third aspect, alone or in combination with one or more of the first and second aspects, the report indicates a delta associated with a CW signal with respect to a reference signal.

[0139] In a fourth aspect, alone or in combination with one or more of the first through third aspects, the report indicates a delta associated with a CW signal with respect to a direct signal.

[0140] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, the report indicates a headroom associated with a CW signal.

[0141] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, the report comprises a power control command.

[0142] Although FIG. **11** shows example blocks of process **1100**, in some aspects, process **1100** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **11**. Additionally, or alternatively, two or more of the blocks of process **1100** may be performed in parallel.

[0143] FIG. **12** is a diagram illustrating an example process **1200** performed, for example, by a transmitting device, in accordance with the present disclosure. Example process **1200** is an example where the transmitting device (e.g., TX device **401** and/or apparatus **1400** of FIG. **14**) performs operations associated with indications and reporting for backscatter reading operations.

[0144] As shown in FIG. 12, in some aspects, process **1200** may include transmitting an indication of an upcoming backscatter reading operation (block **1210**). For example, the transmitting device (e.g., using communication manager **150** and/or transmission component **1404**, depicted in FIG. **14**) may transmit an indication of an upcoming backscatter reading operation, as described herein.

[0145] As further shown in FIG. 12, in some aspects, process **1200** may include transmitting a CW signal to enable the backscatter reading operation (block **1220**). For example, the transmitting device (e.g., using communication manager **150** and/or transmission component **1404**) may transmit a CW signal to enable the backscatter reading operation, as described herein.

[0146] Process **1200** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

[0147] In a first aspect, the indication is transmitted to a receiving device that will perform the backscatter reading operation.

[0148] In a second aspect, alone or in combination with the first aspect, the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

[0149] In a third aspect, alone or in combination with one or more of the first and second aspects, the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

[0150] In a fourth aspect, alone or in combination with one or more of the first through third aspects, process **1200** includes receiving (e.g., using communication manager **150** and/or reception component **1402**, depicted in FIG. **14**) a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0151] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, process **1200** includes receiving (e.g., using communication manager **150** and/or reception component **1402**) a capability message indicating a dynamic range associated with an ADC of a receiving device, such that the indication of the upcoming backscatter reading operation is based on the capability message.

[0152] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process **1200** includes transmitting (e.g., using communication manager **150** and/or transmission component **1404**) a set of beamformed reference signals associated with the upcoming backscatter reading operation, and receiving (e.g., using communication manager **150** and/or reception component **1402**) a report indicating a set of measurements of the set of beamformed reference signals, such that the CW signal is transmitted based on the report.

[0153] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the report comprises a power control command.

[0154] Although FIG. 12 shows example blocks of process **1200**, in some aspects, process **1200** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 12. Additionally, or alternatively, two or more of the blocks of process **1200** may be performed in parallel.

[0155] FIG. 13 is a diagram of an example apparatus **1300** for wireless communication. The apparatus **1300** may be a receiving device, or a receiving device may include the apparatus **1300**. In some aspects, the apparatus **1300** includes a reception component **1302** and a transmission component **1304**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1300** may communicate with another apparatus **1306** (such as a UE, a base station, or another wireless communication device) using the reception component **1302** and the transmission component **1304**. As further shown, the apparatus **1300** may include the communication manager **140**. The communication

manager **140** may include one or more of an adjustment component **1308**, a power management component **1310**, or a measurement component **1312**, among other examples.

[0156] In some aspects, the apparatus **1300** may be configured to perform one or more operations described herein in connection with FIGS. **6-8**. Additionally, or alternatively, the apparatus **1300** may be configured to perform one or more processes described herein, such as process **900** of FIG. **9**, process **1000** of FIG. **10**, process **1100** of FIG. **11**, or a combination thereof. In some aspects, the apparatus **1300** and/or one or more components shown in FIG. **13** may include one or more components of the UE described in connection with FIG. **2**. Additionally, or alternatively, one or more components shown in FIG. **13** may be implemented within one or more components described in connection with FIG. **2**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

[0157] The reception component **1302** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1306**. The reception component **1302** may provide received communications to one or more other components of the apparatus **1300**. In some aspects, the reception component **1302** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1300**. In some aspects, the reception component **1302** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**.

[0158] The transmission component **1304** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1306**. In some aspects, one or more other components of the apparatus **1300** may generate communications and may provide the generated communications to the transmission component **1304** for transmission to the apparatus **1306**. In some aspects, the transmission component **1304** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1306**. In some aspects, the transmission component **1304** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the transmission component **1304** may be co-located with the reception component **1302** in a transceiver.

[0159] In some aspects, the reception component **1302** may receive an indication of an upcoming backscatter reading operation. Accordingly, the adjustment component **1308** may adjust a reception parameter of the apparatus **1300** in response to the indication. In some aspects, the adjustment component **1308** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the transmission component **1304** may transmit a capability message indicating settings associated with an analog passband of the apparatus **1300**, settings associated with an analog stopband of the apparatus **1300**, or a combination thereof. Additionally, or alternatively, the transmission component **1304** may transmit a capability message indicating a dynamic range associated with an ADC of the apparatus **1300**. The reception component **1302** may receive a backscattered signal, using the adjusted reception parameter, based on at least one of an associated priority or an associated QoS.

[0160] Additionally, or alternatively, the reception component **1302** may receive an indication of an

upcoming backscatter reading operation. Accordingly, the power management component **1310** may activate a low-power mode for one or more hardware components of the apparatus **1300** in response to the indication. In some aspects, the power management component **1310** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the adjustment component **1308** may enable an analog passband filter based on one or more desired frequencies and/or enable an analog stopband filter based on one or more continuous wave frequencies.

[0161] Additionally, or alternatively, the measurement component **1312** may measure a set of beamformed reference signals associated with an upcoming backscatter reading operation. In some aspects, the measurement component **1312** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. Accordingly, the transmission component **1304** may transmit a report indicating a set of measurements of the set of beamformed reference signals.

[0162] The number and arrangement of components shown in FIG. 13 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 13. Furthermore, two or more components shown in FIG. 13 may be implemented within a single component, or a single component shown in FIG. 13 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 13 may perform one or more functions described as being performed by another set of components shown in FIG. 13.

[0163] FIG. 14 is a diagram of an example apparatus **1400** for wireless communication. The apparatus **1400** may be a transmitting device, or a transmitting device may include the apparatus **1400**. In some aspects, the apparatus **1400** includes a reception component **1402** and a transmission component **1404**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1400** may communicate with another apparatus **1406** (such as a UE, a base station, or another wireless communication device) using the reception component **1402** and the transmission component **1404**. As further shown, the apparatus **1400** may include the communication manager **150**. The communication manager **150** may include a beamforming component **1408**, among other examples.

[0164] In some aspects, the apparatus **1400** may be configured to perform one or more operations described herein in connection with FIGS. 6-8. Additionally, or alternatively, the apparatus **1400** may be configured to perform one or more processes described herein, such as process **1200** of FIG. 12, or a combination thereof. In some aspects, the apparatus **1400** and/or one or more components shown in FIG. 14 may include one or more components of the network node or of the UE described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 14 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

[0165] The reception component **1402** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1406**. The reception component **1402** may provide received communications to one or more other components of the apparatus **1400**. In some aspects, the reception component **1402** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference

cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1400**. In some aspects, the reception component **1402** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2.

[0166] The transmission component **1404** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1406**. In some aspects, one or more other components of the apparatus **1400** may generate communications and may provide the generated communications to the transmission component **1404** for transmission to the apparatus **1406**. In some aspects, the transmission component **1404** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1406**. In some aspects, the transmission component **1404** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2. In some aspects, the transmission component **1404** may be co-located with the reception component **1402** in a transceiver.

[0167] In some aspects, the transmission component **1404** may transmit an indication of an upcoming backscatter reading operation. Additionally, the transmission component **1404** may transmit a CW signal to enable the backscatter reading operation.

[0168] In some aspects, the reception component **1402** may receive a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof. Additionally, or alternatively, the reception component **1402** may receive a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device.

[0169] In some aspects, the transmission component **1404** may transmit a set of beamformed reference signals associated with the upcoming backscatter reading operation. For example, the beamforming component **1408** may generate control signals to form the set of beamformed reference signals. In some aspects, the beamforming component **1408** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node or of the UE described in connection with FIG. 2. Accordingly, the reception component **1402** may receive a report indicating a set of measurements of the set of beamformed reference signals.

[0170] The number and arrangement of components shown in FIG. 14 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 14. Furthermore, two or more components shown in FIG. 14 may be implemented within a single component, or a single component shown in FIG. 14 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 14 may perform one or more functions described as being performed by another set of components shown in FIG. 14.

[0171] The following provides an overview of some Aspects of the present disclosure:

[0172] Aspect 1: A method of wireless communication performed by a receiving device, comprising: receiving an indication of an upcoming backscatter reading operation; and adjusting a reception parameter of the receiving device in response to the indication.

[0173] Aspect 2: The method of Aspect 1, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

[0174] Aspect 3: The method of Aspect 1, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource

control message.

[0175] Aspect 4: The method of any of Aspects 1 through 3, wherein adjusting the reception parameter comprises: enabling an analog passband filter based on one or more desired frequencies; enabling an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.

[0176] Aspect 5: The method of any of Aspects 1 through 4, further comprising: transmitting a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0177] Aspect 6: The method of any of Aspects 1 through 5, wherein adjusting the reception parameter comprises: adjusting a dynamic range associated with an analog-to-digital converter of the receiving device.

[0178] Aspect 7: The method of any of Aspects 1 through 6, further comprising: transmitting a capability message indicating a dynamic range associated with an analog-to-digital converter of the receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0179] Aspect 8: The method of any of Aspects 1 through 7, further comprising: receiving a backscattered signal using the adjusted reception parameter, wherein the backscattered signal is received based on at least one of an associated priority or an associated quality-of-service.

[0180] Aspect 9: A method of wireless communication performed by a receiving device, comprising: receiving an indication of an upcoming backscatter reading operation; and activating a low-power mode for one or more hardware components of the receiving device in response to the indication.

[0181] Aspect 10: The method of Aspect 9, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

[0182] Aspect 11: The method of Aspect 9, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

[0183] Aspect 12: The method of any of Aspects 9 through 11, wherein the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

[0184] Aspect 13: The method of any of Aspects 9 through 12, further comprising: enabling an analog passband filter based on one or more desired frequencies; enabling an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.

[0185] Aspect 14: A method of wireless communication performed by a receiving device, comprising: measuring a set of beamformed reference signals associated with an upcoming backscatter reading operation; and transmitting a report indicating a set of measurements of the set of beamformed reference signals.

[0186] Aspect 15: The method of Aspect 14, wherein the set of measurements includes one or more reference signal received power values, one or more reference signal received quality values, one or more signal-to-interference-plus-noise ratio values, or a combination thereof.

[0187] Aspect 16: The method of any of Aspects 14 through 15, wherein the report indicates a ratio between a direct signal and a continuous wave signal.

[0188] Aspect 17: The method of any of Aspects 14 through 16, wherein the report indicates a delta associated with a continuous wave signal with respect to a reference signal.

[0189] Aspect 18: The method of any of Aspects 14 through 17, wherein the report indicates a delta associated with a continuous wave signal with respect to a direct signal.

[0190] Aspect 19: The method of any of Aspects 14 through 18, wherein the report indicates a headroom associated with a continuous wave signal.

[0191] Aspect 20: The method of any of Aspects 14 through 19, wherein the report comprises a

power control command.

[0192] Aspect 21: A method of wireless communication performed by a transmitting device, comprising: transmitting an indication of an upcoming backscatter reading operation; and transmitting a continuous wave (CW) signal to enable the backscatter reading operation.

[0193] Aspect 22: The method of Aspect 21, wherein the indication is transmitted to a receiving device that will perform the backscatter reading operation.

[0194] Aspect 23: The method of Aspect 22, wherein the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

[0195] Aspect 24: The method of any of Aspects 21 through 23, wherein the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

[0196] Aspect 25: The method of any of Aspects 21 through 24, further comprising: receiving a capability message indicating settings associated with an analog passband of a receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0197] Aspect 26: The method of any of Aspects 21 through 25, further comprising: receiving a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

[0198] Aspect 27: The method of any of Aspects 21 through 26, further comprising: transmitting a set of beamformed reference signals associated with the upcoming backscatter reading operation; and receiving a report indicating a set of measurements of the set of beamformed reference signals, wherein the CW signal is transmitted based on the report.

[0199] Aspect 28: The method of Aspect 27, wherein the report comprises a power control command.

[0200] Aspect 29: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-8.

[0201] Aspect 30: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-8.

[0202] Aspect 31: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-8.

[0203] Aspect 32: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-8.

[0204] Aspect 33: A non-transitory computer-readable medium storing a set of

[0205] instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-8.

[0206] Aspect 34: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 9-13.

[0207] Aspect 35: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 9-13.

[0208] Aspect 36: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 9-13.

[0209] Aspect 37: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 9-13.

[0210] Aspect 38: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 9-13.

[0211] Aspect 39: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 14-20.

[0212] Aspect 40: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 14-20.

[0213] Aspect 41: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 14-20.

[0214] Aspect 42: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 14-20.

[0215] Aspect 43: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 14-20.

[0216] Aspect 44: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 21-28.

[0217] Aspect 45: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 21-28.

[0218] Aspect 46: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 21-28.

[0219] Aspect 47: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 21-28.

[0220] Aspect 48: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 21-28.

[0221] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

[0222] As used herein, the term “component” is intended to be broadly construed as hardware and/or a combination of hardware and software. “Software” shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a “processor” is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or

methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

[0223] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

[0224] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects.

Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

[0225] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. An apparatus for wireless communication at a receiving device, comprising: a memory; and one or more processors, coupled to the memory, configured to: receive an indication of an upcoming backscatter reading operation; and adjust a reception parameter of the receiving device in response to the indication.
2. The apparatus of claim 1, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.
3. The apparatus of claim 1, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.
4. The apparatus of claim 1, wherein, to adjust the reception parameter, the one or more processors are configured to: enable an analog passband filter based on one or more desired frequencies; enable an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.
5. The apparatus of claim 1, wherein the one or more processors are further configured to: transmit a capability message indicating settings associated with an analog passband of the receiving device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.
6. The apparatus of claim 1, wherein, to adjust the reception parameter, the one or more processors

are configured to: adjust a dynamic range associated with an analog-to-digital converter of the receiving device.

7. The apparatus of claim 1, wherein the one or more processors are further configured to: transmit a capability message indicating a dynamic range associated with an analog-to-digital converter of the receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

8. The apparatus of claim 1, wherein the one or more processors are further configured to: receive a backscattered signal using the adjusted reception parameter, wherein the backscattered signal is received based on at least one of an associated priority or an associated quality-of-service.

9. An apparatus for wireless communication at a receiving device, comprising: a memory; and one or more processors, coupled to the memory, configured to: receive an indication of an upcoming backscatter reading operation; and activate a low-power mode for one or more hardware components of the receiving device in response to the indication.

10. The apparatus of claim 9, wherein the indication is included in downlink control information, a medium access control layer control element, or a radio resource control message.

11. The apparatus of claim 9, wherein the indication is included in sidelink control information, a sidelink medium access control layer control element, or a sidelink radio resource control message.

12. The apparatus of claim 9, wherein the indication identifies one or more time resources, one or more frequency resources, or a combination thereof.

13. The apparatus of claim 9, wherein the one or more processors are further configured to: enable an analog passband filter based on one or more desired frequencies; enable an analog stopband filter based on one or more continuous wave frequencies; or a combination thereof.

14. An apparatus for wireless communication at a receiving device, comprising: a memory; and one or more processors, coupled to the memory, configured to: measure a set of beamformed reference signals associated with an upcoming backscatter reading operation; and transmit a report indicating a set of measurements of the set of beamformed reference signals.

15. The apparatus of claim 14, wherein the set of measurements includes one or more reference signal received power values, one or more reference signal received quality values, one or more signal-to-interference-plus-noise ratio values, or a combination thereof.

16. The apparatus of claim 14, wherein the report indicates a ratio between a direct signal and a continuous wave signal.

17. The apparatus of claim 14, wherein the report indicates a delta associated with a continuous wave signal with respect to a reference signal.

18. The apparatus of claim 14, wherein the report indicates a delta associated with a continuous wave signal with respect to a direct signal.

19. The apparatus of claim 14, wherein the report indicates a headroom associated with a continuous wave signal.

20. The apparatus of claim 14, wherein the report comprises a power control command.

21. An apparatus for wireless communication at a transmitting device, comprising: a memory; and one or more processors, coupled to the memory, configured to: transmit an indication of an upcoming backscatter reading operation; and transmit a continuous wave (CW) signal to enable the backscatter reading operation.

22. The apparatus of claim 21, wherein the indication is transmitted to a receiving device that will perform the backscatter reading operation.

23. The apparatus of claim 22, wherein the receiving device is selected based on a location of the receiving device relative to one or more locations of one or more devices near the receiving device.

24. The apparatus of claim 21, wherein the indication is transmitted to one or more devices near a receiving device that will perform the backscatter reading operation.

25. The apparatus of claim 21, wherein the one or more processors are further configured to: receive a capability message indicating settings associated with an analog passband of a receiving

device, settings associated with an analog stopband of the receiving device, or a combination thereof, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

26. The apparatus of claim 21, wherein the one or more processors are further configured to: receive a capability message indicating a dynamic range associated with an analog-to-digital converter of a receiving device, wherein the indication of the upcoming backscatter reading operation is based on the capability message.

27. The apparatus of claim 21, wherein the one or more processors are further configured to: transmit a set of beamformed reference signals associated with the upcoming backscatter reading operation; and receive a report indicating a set of measurements of the set of beamformed reference signals, wherein the CW signal is transmitted based on the report.

28. The apparatus of claim 27, wherein the report comprises a power control command.
